Kripalu Vipul Sonar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.49
D.R. Highschool & Junior College , Nandurbar	Class XII (Maharashtra State Board)	2024	85.33%
D.R. Highschool & Junior College , Nandurbar	Class X (Maharashtra State Board)	2022	95.20%

SCHOLASTIC ACHIEVEMENTS

International Math Optimization Challenge 2025 : Rank 25 , Srinivasa Ramanujan Mathematics Competition 2025 : Finalist

PSB's FinTech Hackathon 2025 - Finalist, Expedia Group Hackathon 2026 - SemiFinalist, Flipkart Grid 8.0 - SemiFinalist

3x Hackathon Winner : AI Ideathon 2025, Dev Duel 2025, Pulse Quest 2026

Solved **300+** DSA problems across various platforms like LeetCode, Codeforces, CodeChef.

SKILLS

Languages: Python, C++, C

Development: FastAPI, REST APIs, React, Node.js, Flask, Docker, HTML, CSS

Databases: PostgreSQL, MongoDB , MySQL

AI & ML: Machine Learning, Deep Learning, NLP, Generative AI, RAG, Prompt Engineering

Libraries (Python): PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Scikit-learn

Tools & Platforms: Git, GitHub, Linux , VS Code, LaTex, Canva, Microsoft Office Suite, ChatGPT Codex , Claude , Gemini

Concepts: Object-Oriented Programming, Operating Systems, Systems Programming, DSA, DBMS

RELEVANT COURSES

Computer Science : Introduction to Programming, Introduction to AI, Data Structures and Applications, Algorithms, Operating Systems, Database Management Systems (DBMS) , Theory of Computation, Compilers

Mathematics : Discrete Mathematics for Computer Science, Linear Algebra, Probability and Random Variables, Statistics, Transform Techniques, Differential Equations, Calculus, Complex Variables, Numerical Methods, Maths for Physics

PROJECTS

Hotel Review Intelligence Engine (github) :

- Built an AI system analysing 500K+ reviews using aspect-based sentiment, seasonal drift detection, and contradiction handling across traveller types; integrated personalised ranking to deliver context-aware travel recommendations.
- Deployed an interactive dashboard featuring FAISS semantic search, Plotly visualizations, and a multi-criteria

recommendation engine that boosts decision-making through trustworthy, review-grounded evidence and insights.

Numerical Nexus (github):

- Designed and developed a full-stack application featuring interactive user interfaces and robust GUI-based simulations, focusing on excellent user experience and seamless interaction.
- Engineered a scalable FastAPI backend with live computation streaming, applying rigorous troubleshooting and debugging skills to resolve UI layout rendering issues and optimize performance.

Alternative Data Credit Risk Model (github):

- Developed a credit risk assessment system leveraging alternative data sources and machine learning algorithms to predict loan default risk for underbanked populations, enabling scalable and data-driven lending decisions, improving financial inclusion.
- Built a real-time, object-oriented API scoring pipeline with independently testable scoring, data, and explainability layers; debugged and validated model decisions using XAI techniques - National Finalist, PSB FinShield Hackathon.

ChittaRanjanam - Mood-Based Playlist Generator (github) :

- Built an AI-driven music recommendation platform integrating computer vision, emotion recognition models, and Spotify API to generate personalized playlists from user selfies with seamless interaction and real-time emotion analysis capabilities.
- Designed and implemented the ML pipeline for facial emotion detection, mapping user moods to customized music recommendations for an enhanced user experience, personalized content discovery, and intelligent playlist generation.

EXPERIENCE

Tayog - Growth & Product Strategy :

Research collaboration platform connecting students, researchers & institutes across India

- Drove product growth and user engagement strategies through market research, data-driven outreach, and community expansion initiatives across academic networks. Collaborated with cross-functional teams, scaled platform adoption.

POSITIONS OF RESPONSIBILITY

NSS Student Representative(26-27) & Event Coordinator(25-26), IIT Hyderabad :

- Managed volunteer teams to coordinate large-scale social impact initiatives, including 40+ Swachh Bharat drives, civic awareness campaigns, educational programs, orphanage visits, and blood donation camps across the campus.

Inbound Coordinator, International Relations Cell :

- Actively facilitated cultural and academic engagement with international students, enhancing outreach and visibility.

Core Member , Robotix Club :

- Engaged in robotics-focused initiatives and gained exposure to automation, programming, and engineering applications

EXTRACURRICULAR

Continuous Learning: Completed institute-level courses in Japanese and Sanskrit, alongside online certifications in STEM fields to broaden domain knowledge, enhance interdisciplinary learning, and develop diverse technical perspectives

Recipient of Rajya Puraskar (Scouts & Guides), recognizing outstanding teamwork, discipline and community service.

Personal Interests: Indian Classical Music, Chess and staying active with recreational Badminton and Squash.